End-Sem 2024.

(7) $\Box G(x,T) = \delta(x-T) = \delta(x-x')\,\delta(t-t')$.

$$G(x,T) = \frac{1}{(2\pi)^4}\int d^3k \int d\omega\; g(\vec{k},\omega)\; e^{i(\vec{k}\cdot\vec{x} - \omega T)}.$$

[IFT of $g(\vec{k},\omega)$; $\vec{k}$ = wave ~~vector~~ no.]

$$\left(\nabla_x^2 - \frac{\partial^2}{\partial t^2}\right) G(x,T) = \frac{1}{(2\pi)^4}\int dk \int d\omega\; g(k,\omega)\,[(ik)^2 - (-i\omega)^2]\; \exp[\quad]$$

$$= \frac{1}{(2\pi)^4}\int dk \int d\omega\; g(k,\omega)\,[\omega^2 - k^2]\; \exp[\quad]$$

Now, $\displaystyle\int_{-\infty}^{\infty} e^{-i\omega T}\, d\omega = (2\pi)^2\,\delta(T) = (2\pi)^2\,\delta(t-t')$ — (I)

$\displaystyle\int_{-\infty}^{\infty} e^{ikx}\, dk = (2\pi)^2\,\delta(x) = (2\pi)^2\,\delta(x-x')$ — (II)

(IFT)

$\Rightarrow \delta(x-x')\,\delta(t-t') = \delta(x-x')\,\delta(t-t')\cdot g(k,\omega)\,(\omega^2 - k^2)$

[using (I) & (II)]

$\Rightarrow \boxed{g(k,\omega) = \dfrac{1}{\omega^2 - k^2}}$ & (Comparing both sides).

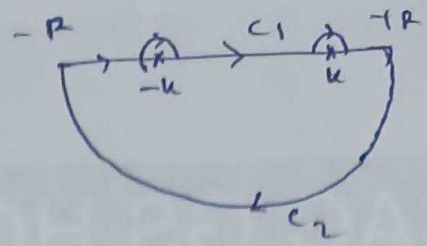
ω- integration:

$$\int_{-\infty}^{\infty} g(k,\omega)\cdot e^{-i\omega T}\, d\omega.$$

$$= \lim_{R\to\infty}\int_{-R}^{R} g(k,\omega)\, e^{-i\omega T}\, d\omega = \oint_C \underbrace{\frac{g\, e^{-i\omega T}}{\omega^2 - k^2}}_{f(\omega)}\, d\omega - \int_{C_2} f(\omega)\, d\omega$$

$\underline{\underline{T > 0}}$: We close C_2 in the lower-half of ϕ.

$\int_{C_2} f(\omega)\, d\omega = 0$ [Jordan's Lemma]



$\oint_C f(\omega)\, d\omega$ has a pole @ $\omega = \pm k$

~~as we are working in lower ½.~~ $C = C_1 + C_2$

[Poles of order = 1].

$\therefore \oint_C f(\omega)\, d\omega = -2\pi i \sum \text{Res}\, f(\omega)\big|_{\omega = \pm k}$ [clockwise direction]

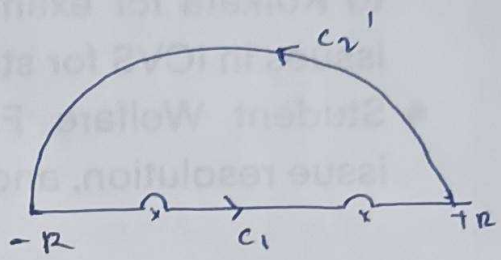
$$= -2\pi i \left[\frac{e^{-i\omega T}}{\omega - k}\Big|_{\omega = -k} + \frac{e^{-i\omega T}}{\omega + k}\Big|_{\omega = k} \right]$$

$$= \frac{-2\pi i}{-2k} \cdot \left(e^{ikT} - e^{-ikT}\right) = \frac{\pi i}{k} \cdot 2i \sin(kT)$$

$$= -\frac{2\pi}{k} \sin(kT)$$

$\underline{T < 0}$: we close C_2' in the upper-half of ½.

$\int_{C_2'} f(\omega)\, d\omega = 0$ [Jordan's Lemma].



$\oint_{C'} f(\omega)\, d\omega = 0$ [no bad points in C']. $C' = C_1 + C_2'$

$$\therefore \int_{-\infty}^{\infty} \frac{e^{-i\omega T}}{\omega^2 - k^2} = \begin{cases} 0 & ; \quad t < t' \\ -\frac{2\pi}{k} \sin(kT) & ; \quad t > t' \end{cases}$$

(Ans)

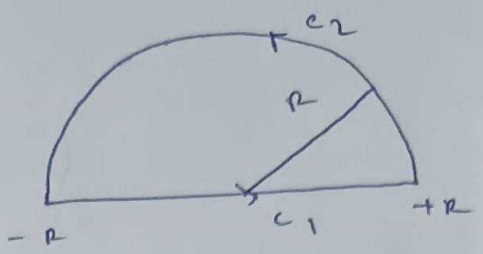
$$\therefore G(x,T) = \begin{cases} 0 & ; \quad t < t' \\ -\frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \frac{dk\, e^{ikx}}{k} \sin(kT) & ; \quad t > t' \end{cases}$$

(Ans)

(3) $$f = \int_0^\infty \frac{\cos x}{q^2+x^2}\,dx$$

$$= \frac{1}{2}\int_{-\infty}^{\infty} \frac{\cos x}{q^2+x^2}\,dx$$

$$= \frac{1}{2}\,\mathrm{Re}\int_{-\infty}^{\infty} \frac{e^{ix}}{q^2+x^2}\,dx$$



$c = c_1 + c_2$

$$= \frac{1}{2}\,\mathrm{Re}\left[\lim_{R\to\infty}\int_{-R}^{R} \frac{e^{ix}}{q^2+x^2}\,dx\right]$$

$$= \frac{1}{2}\,\mathrm{Re}\left[\oint_c \frac{e^{iz}}{q^2+z^2}\,dz - \int_{c_2} \frac{e^{iz}}{q^2+z^2}\,dz\right]$$

$$\int_{c_2} \frac{e^{iz}}{q^2+z^2}\,dz = 0 \quad [\text{Jordan's Lemma}]$$

$$g(z) = \frac{1}{q^2+z^2} \to 0 \quad @ \quad z \to \infty.$$

$\therefore \dfrac{e^{iz}}{q^2+z^2} = f(z)$ has poles @ $z = \pm iq$ of order 1.

Since we are working in the upper-half of $\mathbb{C}$ only.

$z = +iq$ comes inside c.

$$\therefore \oint_c f(z)\,dz = 2\pi i \,\mathrm{Res}\, f(z)\big|_{z=iq}$$

$$= 2\pi i \left.\frac{e^{iz}}{z+iq}\right|_{z=iq} = \boxed{\frac{\pi}{q}\cdot e^{-q}}$$

$$\therefore I = \frac{1}{2}\,\mathrm{Re}\left[\frac{\pi}{q}e^{-q}\right] = \boxed{\frac{\pi}{2q}e^{-q} = I} \quad (\text{Ans})$$

(4) $$g(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(k)\,e^{-ikx}\,dk.$$

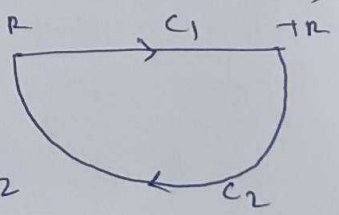
$$= \frac{1}{\sqrt{2\pi}}\boxed{\int_{-\infty}^{\infty} \frac{e^{-ikx}}{(k^2+m^2)^2}\,dk} \longrightarrow R.$$

$$R = \lim_{R\to\infty}\int_{-R}^{R} \frac{e^{-ikx}}{(k^2+m^2)^2}\,dk = \oint_c \underbrace{\frac{e^{-ipz}}{(z^2+m^2)^2}}_{f(z)}\,dz - \int_{c_2} \frac{e^{-ipz}}{(z^2+m^2)^2}\,dz$$

[Renaming $x \to p$ to avoid confusion]

* $\underline{p \geq 0}$:

We close c_2 in the lower half of $\mathbb{C}$.



$$\int_{c_2} f(z)\,dz = 0 \quad [\text{Jordan's Lemma}].$$

$c = c_1 + c_2$

$\frac{e^{-ipz}}{(z^2+m^2)^2} = \frac{e^{-ipz}}{(z+im)^2(z-im)^2}$ has a poles @ $z = \pm im$ of order 2.

Since we are working in lower ½, only $z = -im$ comes inside C.

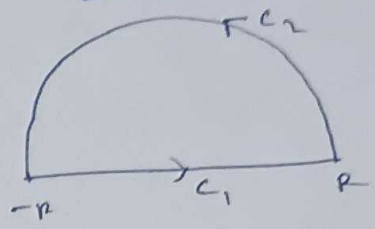
(19)

$$\oint_C f(z)\,dz = -2\pi i \frac{d}{dz}\left(\frac{e^{-ipz}}{(z-im)^2}\right)\Big|_{z=-im} \quad \text{[clockwise direction]}$$

$$= -2\pi i \frac{d}{dz}\left[\frac{(z-im)^2\cdot(-ip) - 2\cancel{(z+im)}}{(z-im)^3}\right] e^{-ipz}\Big|_{z=-im}$$

$$= -2\pi \cancel{i} \frac{d}{d}\left(\frac{+2im(ip) - 2}{8m^3 \cancel{i}}\right)\cdot e^{-pm}$$

$$= -2\frac{\pi}{4m^3}(-2mp-2)\,e^{-pm} = \boxed{+\frac{\pi}{2m^3}(1+mp)e^{-pm}}$$



$C = C_1 + C_2$

$p \leq 0$:

We close C_2 in upper ½.

$$\int_{C_2} f(z)\,dz = 0 \quad \text{[Jordan's lemma]}$$

This time, $z = im$ lies inside C.

$$\oint f(z)\,dz = 2\pi i \frac{d}{dz}\left[\frac{e^{-ipz}}{(z+im)^2}\right]\Big|_{z=im}$$

$$= 2\pi i \cdot \left[\frac{(z+im)(-ip) - 2\cancel{(z+im)}}{(z+im)^3}\right] e^{-ipz}\Big|_{z=im}$$

$$= \frac{\cancel{2}\pi \cancel{i}}{-\cancel{8}_{2}\cancel{i}m^3}\left[\cancel{2}im(-ip) - \cancel{2}\right] e^{pm}$$

$$= \boxed{\frac{\pi}{2m^3}(1-mp)\,e^{pm}}$$

$\therefore$ $p \geq 0 \longrightarrow I = \frac{\pi}{2m^3}(1+mp)\,e^{-pm}$

$p < 0 \longrightarrow I = \frac{\pi}{2m^3}(1-mp)\,e^{pm}$

Combining the 2: $I = \frac{\pi}{2m^3}(1+m|p|)\,e^{-p|m|}$

$$\cancel{g(x) =}\ \boxed{I = \frac{\pi}{2m^3}(1+m|x|)\cdot e^{-x|m|}}$$

$$\boxed{g(x) = \frac{1}{\sqrt{2\pi}} I} \quad \text{(Ans)}$$

(5) $P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n}(x^2-1)^n$ —— (I) (20)

(i) From Cauchy's Integral Formula:

$$f^n(z_0) = \frac{n!}{2\pi i}\oint_C \frac{f(z)}{(z-z_0)^{n+1}}\,dz \;;$$

C is a contour encircling z_0 in anti-clockwise direct^n.

Let $f(z') = (z'^2-1)^n$ —— (II) ; $z_0 = z$; $z \to z'$

$$\therefore f^n(z) = \frac{n!}{2\pi i}\oint_C \frac{f(z')}{(z'-z)^{n+1}}\,dz'$$

$$\cancel{\Rightarrow \frac{d^n}{dx^n}x'^2 = \frac{n!}{2\pi i}\oint_C}$$

$$\Rightarrow \frac{d^n}{dz^n}(z^2-1)^n = 2^n n!\, P_n(z) \quad \text{[from (I)]}.$$

$$\Rightarrow \boxed{P_n(z) = \frac{1}{2^n 2\pi i}\oint_C \frac{(z'^2-1)^n}{(z'-z)^{n+1}}\,dz'} \quad \text{[from (II)]}.$$

(Proved)

(ii) @ $z=1$:

$$P_n(1) = \frac{1}{2^n 2\pi i}\oint_C \frac{(z'^2-1)^n}{(z'-1)^{n+1}}\,dz' \to f(z')$$

$$\cancel{\frac{(z'^2-1)^n}{(z'-1)^{n+1}} = \frac{(z'-1)^n(z'+1)^n}{(z'-1)^{n+1}} = \frac{(z'+1)^n}{(z'-1)}\,dz'}$$

$\cancel{@\ z'=1\ ;\ f(z')}$ $\cancel{\text{has a pole of order } (n+1).}$

$$\cancel{\therefore \oint_C f(z')\,dz' = 2\pi i \operatorname{Res} f(z')\big|_{z'=1}}$$

$$\cancel{= 2\pi i \frac{1}{n!}\frac{d^n}{dz^n}(z'^2-1)^n\Big|_{z'=1}}$$

$$\cancel{= \frac{2\pi i}{n!}\, n\cdot(n-1)(n-2)\ldots 2^n}$$

$$\frac{(z'^2-1)^n}{(z'-1)^{n+1}} = \frac{(z'^2+1)^n}{(z'-1)}.$$ $f(z)$ has a simple pole @ $z'=1$.

$$\therefore \oint_C f(z')\,dz' = 2\pi i \operatorname{Res} f(z')\big|_{z'=1} = 2\pi i\cdot(z'^2+1)^n\big|_{z'=1} = \boxed{2\pi i\cdot 2^n}$$

$\therefore P_n(1) = \frac{1}{2^n 2\pi i} \cdot 2\pi i \cdot 2^n = 1$. (Proved)

6) For a unitary operator: $U^\dagger U = UU^\dagger = \mathbb{I}$.

Let $|v\rangle$ & $|w\rangle$ be 2 eigenvectors of U.

(i) — $U|v\rangle = \lambda_1 |v\rangle$. [$\lambda_1, \lambda_2$ – eigenvalues].

$U|w\rangle = \lambda_2 |w\rangle$

$\Rightarrow \langle w| U^\dagger = \langle w| \lambda_2^*$ — (ii)

From (i) & (ii): $\langle w| U^\dagger U |v\rangle = \langle w|v\rangle \lambda_1 \lambda_2^*$

$\Rightarrow \langle w|v\rangle = \lambda_1 \lambda_2^* \langle w|v\rangle$

$\Rightarrow \boxed{\langle w|v\rangle (1 - \lambda_1 \lambda_2^*) = 0}$ — (iii)

~~Now~~, From (i): $\langle v| U^\dagger = \langle v| \lambda_1^*$

[taking Hermitian conjugate of (i)]. $\Rightarrow \langle v| U^\dagger U |v\rangle = \langle v|v\rangle |\lambda_1|^2$

$\Rightarrow \langle v|v\rangle = \langle v|v\rangle |\lambda_1|^2$ [$\langle v|v\rangle = |v|^2 \neq 0$]

$\Rightarrow \boxed{|\lambda_1|^2 = 1} \Rightarrow \boxed{|\lambda_1| = 1}$

~~Si~~ Similarly, $|\lambda_2|^2 = 1$. (unitary operators have unit modulus eigenvalues)

from ~~(ii)~~ (iii); either $1 - \lambda_1 \lambda_2^* = 0$.

$\Rightarrow \lambda_1 \lambda_2^* = 1$.

$\Rightarrow \boxed{\lambda_1 = \frac{1}{\lambda_2^*}}$ — (iv)

Let λ_1, λ_2 be a complex no.; $\lambda_1 = a_1 + ib_1$ ~~& $|\lambda| = 1$~~.

~~$\lambda_2 = a_2 + ib_2$~~

$a_1^2 + b_1^2 = 1$

~~$a_2^2 + b_2^2 = 1$.~~

$\lambda_1^* = a_1 - ib_1 \Rightarrow \frac{1}{\lambda_1^*} = \frac{1}{a_1 - ib_1} = \frac{a_1 + ib_1}{a_1^2 + b_1^2} = a_1 + ib_1 = \lambda_1$.

$\therefore \boxed{\lambda_1^* = \frac{1}{\lambda_1}}$ & same for λ_2.

$\therefore$ (iv) implies $\lambda_1 = \lambda_2$ ~~$\Rightarrow$ Not pos~~

2nd possibility from (iii) $\Rightarrow \boxed{\langle w|v\rangle = 0}$ – Orthogonal.

Thus, eigenvectors corresponding to different eigenvalues of a unitary operator are orthogonal.